

Artificial intelligence and ChatGPT are fostering knowledge sharing, ethics, academia and libraries

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Abstract

Purpose – Given the increasing attention on ChatGPT in academia due to its advanced features and capabilities, this study aims to examine the links among Artificial intelligence (AI), knowledge sharing, ethics, academia and libraries in educational institutions. Moreover, this study also aims to provide a literature base while discussing recent trends in AI and ChatGPT technologies, highlighting their specific uses in institutions.

Design/methodology/approach – The paper involves a structured interview format where a human interviewer poses questions “Qs” in ChatGPT, related to knowledge sharing, ethics, academia and libraries. Moreover a literature base is also provide to discussed recent trends in AI and ChatGPT technologies, highlighting their specific uses in institutions.

Findings – The study find out that AI and ChatGPT technologies in educational institutions affect knowledge sharing, ethical consideration, academia and libraries. This study also highlights literature directions for the trends and proper use of the AI and ChatGPT among institutions, such as improving student-learning engagement.

Originality/value – This research contributes to the prior literature by offering an in-depth review of current uses and applications of AI and ChatGPT in educational institutions. It not only highlights key trends and innovations but also provides insights and guidelines for future research. This study also provides insights and guidelines for future research. Furthermore, the article emphasizes the potential impact of AI and ChatGPT on the future of education and technology.

Keywords Artificial intelligence, ChatGPT, Knowledge sharing, Ethics, Academia, Libraries, Institutions

Paper type Research paper

1. Introduction

ChatGPT is an AI-powered chatbot developed by OpenAI that employs GPT language model technology. It is accessible to the public and can respond to various text-based queries, from generating appreciation notes to offering advice on different problems and providing innovative solutions. ChatGPT has access to a vast collection of data and advanced architecture, enabling it to understand and interpret user inquiries and offer responses that closely resemble natural human language. Taecharungroj (2023) stated that “ChatGPT” can produce language” that emulates human speech and can perform complex tasks, marking a significant advancement in the fields of Natural Language Processing (NLP) and AI.

The author(s) utilized ChatGPT to aid in constructing well-structured sentences and ensuring grammatical accuracy.



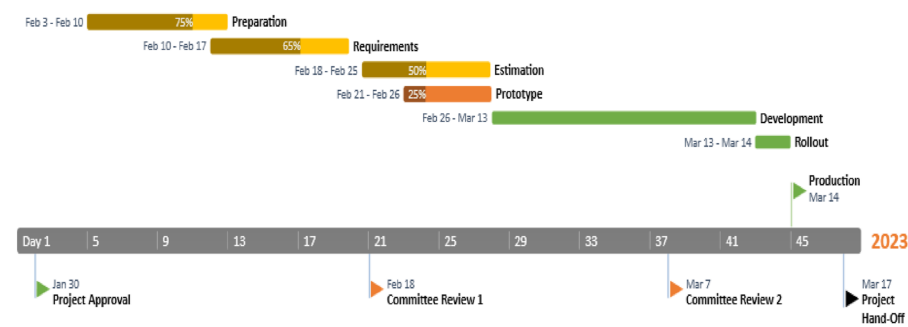
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Specifically, a research lab was established by OpenAI in 2015 as noted by [Brockman et al. \(2016\)](#), achieving remarkable milestones such as DALL-E and ChatGPT, both of which are now publicly accessible. DALL-E, introduced by [Devlin et al. \(2018\)](#), is a machine learning tool that generates distinct images based on user input using natural language processing. The broad accessibility of DALL-E played a central role in the rapid rise in popularity of ChatGPT, which surpassed one million unique users within just one week of its launch. OpenAI has developed a language model called ChatGPT, capable of generating text responses that closely resemble human language ([Dale, 2021](#); [Borji, 2023](#); [Dergaa et al., 2023](#); [Kocoń et al., 2023](#)). Specifically, the GPT method comprises two stages: unsupervised pretraining using unlabeled data to develop underlying concepts, followed by supervised fine-tuning to enhance task-specific performance, as discussed in previous studies.

Cutting-edge language models such like ChatGPT and GPT-3 have been developed by OpenAI, Google and Microsoft. Each model possesses distinct capabilities and can be employed for various natural language processing tasks. GPT-3 and ChatGPT are well-suited for purposes such as summarization, language translation and answering various types of queries, as they can produce text that closely resembles human writing. Meanwhile, Bidirectional Encoder Representations (BERT), XLNet excel and RoBERTa excel at evaluating the meaning of text, making them useful for specific tasks. While the primary advantage of ChatGPT and GPT-3 is their extraordinary capability to generate high-quality text. On the other hand, BERT and XLNet are most suitable for analyzing and comprehending text. The timeline of OpenAI’s developmental portfolio is represented in [Figures 1–3](#).

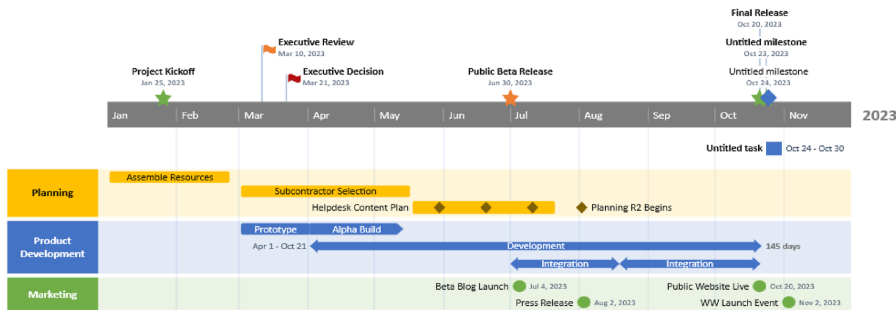
OpenAI introduced the advanced chatbot tool, ChatGPT, which performs a variety of assigned text-based tasks. These tasks range from answering simple to complex questions generating academic essays, and more. ChatGPT can break down a main essay into subtopics and write each section using GPT, enabling the production of a complete article with minimal researcher input. Moreover, natural language processing is an advanced tool in ChatGPT, useful for handling basic customer service inquiries, assisting with legal research and document preparation and overseeing the quality of written work in institutions.

While, GPT technology hold great potential for natural language processing applications, it also has some disadvantages. As noted by [Dale \(2021\)](#) and [Lucy and Bamman \(2021\)](#), GPT models are generated through a statistical method that identifies patterns in a vast collection of text data, which can lead to the reinforcement of biases and prejudices present in the data. These are the main limitations of the GPT models. This highlights the potential for the model to produce output that may be detrimental or unsuitable. Furthermore, while GPT technology is proficient at generating text, it lacks full comprehension and understanding of the context and significance of the generated content.



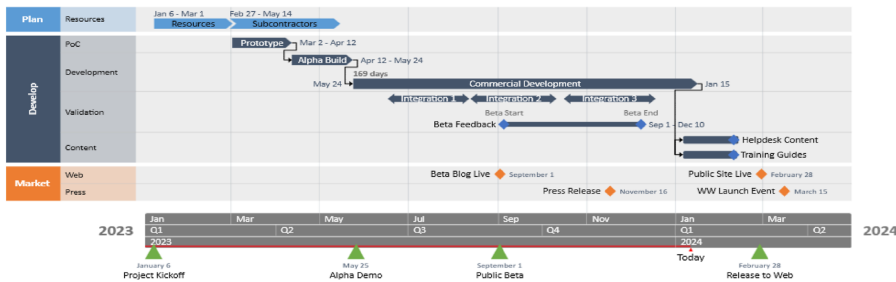
Source(s): OpenAI ChatGPT Timeline (Openai.com)

Figure 1. Project portfolio of OpenAI ChatGPT timeline



Source(s): OpenAI ChatGPT Timeline (Openai.com)

Figure 2. Project portfolio of OpenAI ChatGPT timeline



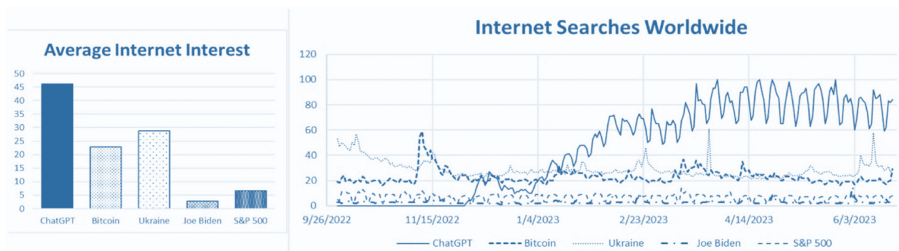
Source(s): OpenAI ChatGPT Timeline (Openai.com)

Figure 3. Project portfolio of OpenAI ChatGPT timeline

These models struggle with tasks requiring common sense or logical reasoning, as such aspects are not included in their training data. Training GPT models requires substantial computational power and large quantities of data, which can pose challenges for certain individuals and organizations. This model can pose obstacles for specific individuals and organizations. Additionally, it is important to recognize the significant energy consumption required to operate the algorithms and data centers at the scale of OpenAI, as noted by [Zhou et al. \(2020\)](#), [Mahbod et al. \(2022\)](#) and [Hongdae and Pasricha \(2024\)](#).

To evaluate the media's influence following the introduction of ChatGPT on November 30, 2022, an analysis of Google user search trends using Google Trends was conducted. This analysis illustrates the search volume of queries over time across various countries and languages (Refer to [Figure 4](#)), demonstrate the significant impact of ChatGPT. Since its launch in November 2022. The interest in AI-based application exceeded that of topics such as the Ukraine war, news related to U.S. President Joe Biden, Bitcoin and the S&P 500. The data depicted in the chart corresponds with the findings of Libert, indicating a remarkable surge of 112,740% in searching interest for ChatGPT respectively.

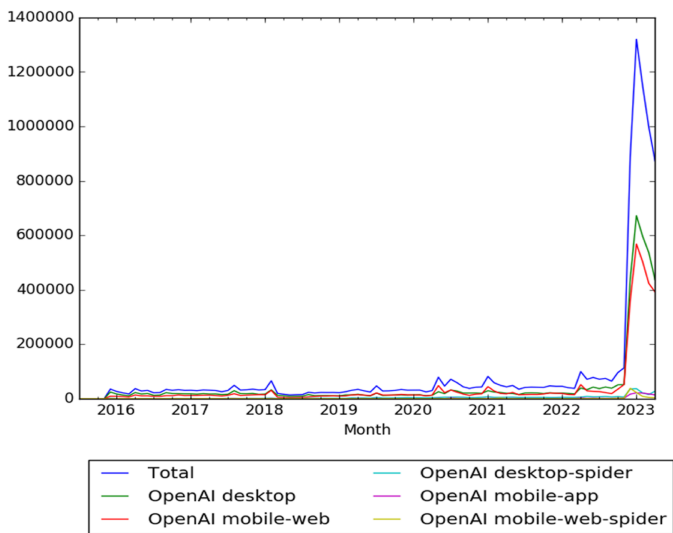
The below (refer to [Figure 5](#)) displays the number of views on the OpenAI Wikipedia page in English language which was started from July 2016 to March 2023. The following segment discusses the significance of AI and ChatGPT in academic institutions. Also introduced along with certain fundamental ideas regarding ChatGPT, such as natural language processing and AI, and their function in technology. The different terminologies and concept of ChatGPT are presented in (refer to [Table 1](#)). The remaining part of the subsequent section presents literature review, methods, results and discussion respectively.



Note(s): The figures indicate search interest compared to the maximum point on the graph for the specified area and duration. A score of 100 signifies the zenith of popularity for the phrase. A rating of 50 implies the word is only half as prevalent. A value of 0 indicates insufficient data for the given term

Source(s): Google trends (2023)

Figure 4. Search interest based on Google trends



Source(s): OpenAI ChatGPT Timeline (Openai.com)

Figure 5. Number of views on the OpenAI

2. Literature review

To cope worldwide educational challenges, institutions must consistently adopt innovative strategies. These challenges are driven by factors such as government funding, increasing internationalization and a growing online education market. Qiping and White (2023) research on the education challenges reveals that educational institutions must balance their traditional societal role of producing skilled workers with treating students like paying customers who expect high-quality services. Mainardes *et al.* suggest that in order to evaluate sources of competitive advantage and challenges, it is important to examine resources and abilities and the role of stakeholders. Physical infrastructure and technology are referred to as resources and abilities. Which are essential for competitive advantage and addressing challenges. While the external factors, such as government regulations and potential risks from online educational

Table 1. Key concepts related to ChatGPT

Concept/Source	Statements
Text generation Haleem et al. (2023)	Text generation is a ChatGPT language model. This model has a massive amounts of textual data and has learned to generate coherent and grammatically correct text
Deep learning Goh et al. (2017)	Deep learning is an algorithms technique. It useable for wide range of tasks, including natural language processing, speech recognition and image recognition. This tools particularly effective for tasks where traditional rule-based programming or statistical modeling techniques are inadequate
Attention mechanisms Bilal et al. (2022)	Attention mechanisms are used for numerous modern neural network architectures. This network system focus on the most relevant parts of the input data, rather than treating all parts equally
GPT Lecler et al. (2023)	GPT models are used for an unsupervised learning approach. This model learns to predict the next word in a sequence based on the words that came before it in the training data, without being explicitly told what the correct answer is
Chatbot Adamopoulou and Moussiades (2020)	Chatbot is a computer program. This program designed to simulate conversation with human users, typically via voice input or text. Chatbots program perform various tasks, such as providing customer support or answering frequently asked questions
Generative models Kumar and Sinha (2023)	Generative models is a machine learning model. This model is designed to generate new data that is similar to the input data that it was trained on
Language model Sun and Hoelscher (2023)	A language model generate text that is coherent and contextually appropriate. This ChatGPT language model is able to take into account the structure of language, such as grammar, syntax and semantics, and it can generate responses that are appropriate to the specific context
Multimodal neurons Wang et al. (2023)	Multimodal neurons are thoughts to play an important role in perception and cognition. Multimodal neurons allow the brain to integrate information from different sensory modalities to create a more holistic understanding of the world. This is important for tasks that require the integration of sensory information, such as social communication, navigation and object recognition
Neural networks Bishop (1994)	Neural networks is applicable for a variety of applications. These applications including image and speech recognition, natural language processing and autonomous vehicles. Neural networks demonstrated impressive performance on a variety of tasks, particularly when trained on large amounts of data
Natural language processing Pruneski et al. (2022)	Natural language processing is developing algorithms and computer programs that assist machines to understand, interpret and generate human language
Transfer learning Torrey and Shavlik (2010)	Transfer learning is a machine learning technique. In which a pre-trained model is used as a starting point for a new task, rather than training a new model from scratch. Transfer learning is become progressively popular in recent years. This used particularly in deep learning applications, such as computer vision and natural language processing
Supervised fine-tuning Ali et al. (2017)	Supervised fine-tuning is a machine-learning technique. It involves taking a pre-trained model and further training it on a specific task using labeled data. In addition, used in natural language processing tasks, such as sentiment analysis or language translation
Unsupervised pretraining Ciga et al. (2022)	Unsupervised pretraining is a machine learning technique. This technique involves training a model on a large dataset without the use of labels or supervision. The unsupervised pretraining is a meaningful representation of the input data, which starting point for supervised learning on a smaller labeled dataset
Source(s): Own source	

institutions are also important for cover challenges and sustain competitive advantage. Additionally, it is important to consider the prominent role of stakeholders, such as faculty, students and society. With the advent of new information and communication technologies, the educational setting has undergone significant changes, leading to increased competitive

pressures (Zeb *et al.*, 2023). Despite the importance of certain elements in competition and challenges breakdown, limited research has been conducted on their impact. This study aims to extensively investigate the connections between cutting-edge technology and various factors related to learning. The adoption of AI and ChatGPT3 in all aspects of operations will enable organizations to maintain a competitive edge and cope challenges. This trend progresses toward greater integration of technology and education, and in turn, meet the expectations of their stakeholders in easy way.

According to experts in the fields of education and training, AI refers to computer systems capable of engaging in human-like activities such as learning, self-correction, synthesizing, adapting and performing complex processing tasks using data (Alimi *et al.*, 2021; Chávez *et al.*, 2023). However, Bearman and Ajjawi (2023), impact of AI on education, learning, research and assessment is still not fully understood and remains a complex and ambiguous topic. The Generative Pre-Trained Transformer (GPT-3), Dale (2021), Ghajargar *et al.* (2022) and Zhang *et al.* (2023), is an AI technology, that has made significant progress and capable of producing written language that closely resembles like a human, with or without user input. One notable example of this technology is the ability of AI to completely rephrase sentences within commonly used software (Montemayor, 2021).

The advancement in AI technology offer the possibility of increasing the accessibility and inclusivity of content through various tools like text summarization, live captioning, machine translation and pre-existing collections of idioms and phrases (Strobelt *et al.*, 2021). Rao and Meo (2016), and Roski *et al.* (2021), Universal Design for Learning model defines technology as a group of tools that aid different forms of communication, illustration and articulation. Therefore the extensive use of AI in educations institutions are confronted with a complicated reality today. The prevalence of AI technology in education has created a complex situation for educators. It is likely that many educationists are not adequately equipped to handle these challenges that arise from the use of AI in higher education (Zhang and Aslan, 2021; Luan *et al.*, 2020). To effectively handle these issues, it is necessary to have a wide-ranging knowledge of how to navigate the intricacies of AI. Brown (2015) claimed that it is important to distinguish between humans and AI, and to discern between AI tools that provide support versus those that enable duplicitous.

Up-to-date trends between researchers and AI has been increasing day by day. As the researchers have underscored the importance of prioritizing human capabilities like critical thinking, problem-solving and questioning and answering in the face of AI advancements (Eguchi *et al.*, 2021; Jaiswal *et al.*, 2022; Dwivedi *et al.*, 2023; Kasneci *et al.*, 2023). This highlights the importance for academic discourse concerning AI in higher education to start a course for future progress. A pressing concern within contemporary higher education is how the integration of AI might affect academic integrity (Velliariis and Breen, 2016). The aforementioned statements clarify the ethical and honest use of AI in educational institutions.

AI has the potential to transform higher education by enhancing the quality of teaching and learning activities, research capabilities and improving administrative efficiency (Hannan and Liu, 2023; Zeb *et al.*, 2024; Ullah *et al.*, 2024). They further studies the various applications of AI in educational institutions, including intelligent tutoring systems, predictive analytics and chatbots. These researchers investigate the difficulties and moral factors linked with the implementation of AI technology in educational institutions. They further claimed that AI provide beneficial possibilities for education establishments to maintain their competitiveness. Therefore, it is important to be thoughtfully consider the ethical consequences and the potential effect on human abilities and employment. Finally, careful consideration is necessary before implementing AI in educational institutions.

The proper use of Generative AI in knowledge management has significant implications as it involves the accumulation, organization, examination and dissemination of knowledge within education institutions. This AI innovative technology brings to light new perspectives introduces novel possibilities for systematic investigations. Current research reveals that Generative AI can be applied in knowledge management for various purposes such as

retrieving information by [Guo et al. \(2020\)](#), mining texts by [Kumar et al. \(2021\)](#), identifying topics by [Mustak et al.](#) and personalizing e-learning environments by [Arun Kumar et al. \(2022\)](#). The institutions must prioritize ethical concerns to uphold their reputation as reliable sources of information and guarantee that the utilization of ChatGPT is consistent with their principles and objectives ([Lund and Wang, 2023](#); [Ray, 2023](#); [Dwivedi et al., 2023](#); [Rasul et al., 2023](#)).

The reactions from the scientific and academic communities towards ChatGPT have been varied, emphasizing the ongoing debate surrounding the advantages and disadvantages of cutting-edge AI technologies ([Wogu et al., 2017](#); [Tai, 2020](#); [Bozkurt et al., 2023](#)). On one hand, supporters argue that ChatGPT and similar large language models offer significant advantages, particularly in improving conversational interactions and aiding in writing tasks, thereby improving the efficiency and accuracy of desired outcomes ([Deng and Lin, 2022](#)). On another hand, some claims have been raised regarding potential biases that could limit its capabilities and lead to the generation dissemination of factually wrong information. However, it is essential to address security prospective, including the potential risk of cyber-attacks facilitated by the distribution of non-authentic information through Language Model Systems ([Deng and Lin, 2022](#); [Yao et al., 2024](#)). However, higher education institutions utilize AI, but ethical concern is necessary for AI particularly in the future scenarios ([Zawacki-Richter et al., 2019](#); [Ullah et al., 2024](#)). The main objective of this study is to examine the links between AI and ChatGPT on knowledge sharing, ethical considerations, academia and libraries. Additionally, the study will highlight literature on the trends and useful applications of AI and ChatGPT technologies in educational institutions. The subsequent sections of this review study are presented as follows: method, results and discussion.

3. Methodology

In this study, we utilized the ChatGPT-4 architecture for the questionnaire experiment. This description ensures reproducibility and clarity of our study. Moreover, allowing others to replicate the study with the same version of the model. The methodology involves a structured interview format where a human interviewer poses questions “Qs” in ChatGPT, related to knowledge sharing, ethics, academia and libraries and the AI’s responses are analyzed. The structure of the human interview approach in ChatGPT is a novel approach ([Tülübaş et al., 2023](#); [Barambones et al., 2024](#)). This approach is straightly ties into the research objectives of exploring the connections between AI, knowledge sharing, ethics, academia and libraries. The given declarations provide information that how it aligns and justifies the research objective. Firstly, ChatGPT is an AI model, making it a fitting subject for exploring the implications of AI in knowledge sharing and academia. Moreover, the use of ChatGPT as the interviewee allows for direct interrogation of the technology’s capabilities, potential applications and limitations in these context. This bring into line with the purpose of investigating AI’s role in knowledge sharing and academic settings. Secondly, the used of specific ethical questions in ChatGPT examines AI’s ability to handle ethical concerns. This comprises its capacity to address issues such as data privacy such as transparency and bias. The format allows us to assess the model’s responses in line with ethical backgrounds, fulfilling the objective of exploring responsible AI usage. Thirdly, the interview on libraries and their evolving role with AI. This study apprehensions the intersection of traditional knowledge repositories with the cutting-edge technology usage. This approach for a meaningful investigation into how AI can transform knowledge management and dissemination in academic libraries, further aligning with the research purpose.

The questions were related to knowledge sharing, ethics, academia and libraries are posed to ChatGPT by a human interviewer, to gather insights. We have drawn parallels between interviewee answers and key themes in the literature, thereby reinforcing the study’s objectives and contributions. Each question is properly aligned with research objectives, ensuring a comprehensive exploration of these themes. Consequently, the academia section focuses on

the role of AI in educational settings such as knowledge sharing and libraries. While the ethics section probes into responsible AI usage and data privacy concerns. Lastly, a prior literature base was also provided discussing recent trends in the application of these technologies highlighting their specific uses in academia, and formulating conclusion about their future scope.

4. Results

The answers/responses to these questions are exclusively generated by ChatGPT, labeled as “Ans”. This technique allows for exploring various facets of AI and ChatGPT’s impact on these domains directly from the perspective of AI system itself. The results of questionings and answering are presented in (refer to [Table 2](#)). Additionally, a comprehensive literature background has provided which is discussed recent trends in AI and ChatGPT technologies, highlighting their specific uses in institutions.

5. Discussion

The introductory section of this study briefly presents a comprehensive overview of ChatGPT, and AI-based chatboat has garnered significant attention in the education sector due to its potential to enhance competitiveness and support educators. The initial part of the manuscript is dedicated to exploring the inception, functionalities and framework of GPT, and the foundational algorithm upon which ChatGPT is built. It describes how GPT can perform various language-related tasks and explains key concepts closely associated with ChatGPT. Furthermore, the study elaborates on various advantages of ChatGPT, including but not limited to reference provisioning, metadata generation, content generation facilitation and discovery enhancement. While the second part mainly focuses on providing a transcript of an interview discussing the subjects of knowledge sharing, ethics, academia, libraries and GPT as they to the main related to the main objectives of the current study.

The findings revealed that ChatGPT3 has the capability to enhance the efficacy and precision of information dissemination. It can be utilized, for instance, to develop chatbots and virtual assistance to provide information on a variety of subjects and responding to frequently asked inquiries. This can be especially beneficial for academic institutions and libraries that must provide information access to a large number of individuals. The finding is consistent with previous studies that ChatGPT have a prominent role in knowledge sharing, collaboration and learning in the digital workplace context ([Fahad et al., 2024](#); [Sumbal and Amber, 2024](#)). The findings also concluded that ethical considerations must be taken into account, including privacy and bias. The results is tie with the findings of [Dave et al. \(2023\)](#) and [Stahl and Eke \(2024\)](#). They stated that the potential use of ChatGPT and other AI tools must poses ethical and legal concerns. These include possible copyright laws, legal complications and other needed transparency in AI-generated content. Furthermore, ethical ramifications of utilizing AI language-based models for knowledge sharing, such as ChatGPT3, must be taken into account for proper usage in academia. Such models can run the risks of perpetuating discrimination and biases that are inherent in the data on which they are trained. Consequently, it is critical to monitor and assess the output of these models with extreme care to ensure that they do not propagate false information or detrimental stereotypes.

The results of the current study further revealed that academics might utilize ChatGPT3 for writing and research assistance. For instance, it can help researchers locate pertinent literature on a given subject or generate summaries of scholarly articles. The statement is further supported by [Bin-Nashwan et al. \(2023\)](#), it is critical to guarantee any utilization of ChatGPT3 in scholarly investigations is unambiguous and conspicuously cited in any resultant publications. However, it is essential to avoid using exact information from articles due to similarity issues, and the references have not usually properly generated. The finding of the current study lastly concluded that ChatGPT3 could be utilized in libraries to assist users in

Table 2. Questions with ChatGPT about its Impact

Particular	Questions and answer
Knowledge sharing	Qs; How does ChatGPT impact knowledge sharing? With ability to generate human like text and providing the authentic answers within seconds on a wider scale. Will the AI is directly affecting the capabilities of human beings and it is reducing efficiency of human beings? Ans; As an AI is empowered tool to generate the content within seconds according to the requirement of every human being, which is ultimately sharing knowledge and data with individuals. Which is also making easy the requirement of individuals But at the same time the concept of knowledge sharing and learning is reducing due to having all the information on single click. Which is effecting efficiency of learning new things among individual is reducing Overall impact of knowledge sharing and AI generated information is based upon the use of human being that how human being is using the ChatGPT in their work place or for educational purposes. At the same time there is risk of using AI for negative purposes. Which may also cause a big issue
Ethical considerations	Qs; What are the ethical considerations of using ChatGPT? How we can ensure to use the information created by ChatGPT will not create any misinformation, which will ultimately effect human beings as well as how we can ensure cyber security while the AI tool is generating same content as human beings? Ans; ChatGPT is a robust resource and there are certain tools in it which is securing to generate information contains misinformation and propaganda. However, at the same time human beings are surrounded with feelings and emotions, so it is our responsibility to use AI tools with responsibility At the same time, it's also important to add programs in ChatGPT which leads to secure privacy of human beings and reduce the risk of generating misinformation which may leads to the generating propaganda Transparency and accountability are two important concepts while using ChatGPT that the information generated in ChatGPT is based upon true data To ensure using ChatGPT in responsible way it is important to set clear guidelines, rules, regulations and code of conducts Finally, accountability can ensure the use of ChatGPT and imposing penalties and fines on companies and people who are using ChatGPT in unethical way
Academia	Qs; What impact does ChatGPT have on academia? Is the ChatGPT is effecting academia that how people use to conduct research, publishing articles and doing their projects? Alternatively, does it creating a new means of creating information, which is helping individuals to achieve their academic goals? Ans; ChatGPT is assisting researchers to conduct research and working on large end data to concise it in simple way ChatGPT can also help individuals to work on their concern research topics and provide them authentic data in meaningful form ChatGPT can provide individuals with the platform to publish their research and connect with billions of individuals ChatGPT may also affect the traditional publishing industry by adding innovations and changes to make the work more creative Finally, ChatGPT can create new opportunities for knowledge spreading and collaboration likewise bring researchers and academia professionals on a single page

(continued)

Table 2. Continued

Particular	Questions and answer
Libraries	QS; How can libraries leverage ChatGPT to enhance their services? Can ChatGPT gather data and information from libraries to enhance its knowledge and information more fact based? Ans; Libraries can leverage ChatGPT in several ways to enhance their services and authenticity ChatGPT can use several information from libraries for adding references and accommodation to its information On the other way, libraries can use ChatGPT to give quick responses to the questions and providing accurate answers. Which may leads to bring virtual AI tools in libraries ChatGPT can also help users in library to provide them recommendation to read books based upon their previous history, like what they read in the past At the same time, ChatGPT can also help in libraries to bring natural language and modern means of information To ensure that the use of AI language models aligns with the mission and goals of libraries they should implement certain rules and polices to ensure the safety of information generated by AI tools. These policies should be developed in collaboration with stakeholders, including library staff, users and experts in AI ethics Overall, libraries can leverage ChatGPT to enhance their services and provide information, which a user can use in most efficient and effective ways. However, it is important to ensure that the use of AI language models aligns with the library's mission and values, which may lead to secure the ethical standards of library As the users of ChatGPT are more reliable on technology, students can use AI tools for the significant impact on their education. However, at the same time it is essential to set limitation to reduce the chances to over relaying on AI tool

Source(s): Own source

locating pertinent books and other resources and delivering support for research inquiries. It is ties with the results of [Adetayo \(2023\)](#) and [Lund et al. \(2024\)](#). They endorsed that it is critical to ensure that any library implementation of ChatGPT3 is conducted in an ethical and transparent manner, with due consideration given to any potential biases in the training data. While highlighting the potential benefits of ChatGPT for librarianship and academic research, this article also issues admonitions regarding potential drawbacks. Hence, in order to advance our research and education, it is vital that we determine how to collaborate with this technology in a responsible and ethical manner.

5.1 Implications

The study underscores the advantages AI and ChatGPT to facilitate better knowledge sharing among students and faculty. ChatGPT enables better knowledge sharing. In addition, it is acting as a virtual assistant to help manage tasks, schedules and other activities. ChatGPT enables rapid and accessible knowledge sharing by providing immediate responses to queries across a wide range of topics ([Nazir and Wang, 2023](#)). This can democratize access to information, allowing people from various backgrounds to benefit from expert-level knowledge on various topics. Moreover, this study underscores the importance of addressing ethical concerns. ChatGPT raises key ethical considerations regarding the usage of AI in generating content ([Uzun, 2023](#)). It highlights the need for responsible AI development, ensuring transparency, addressing biases and encouraging ethical guidelines for AI usage. It further stresses the safety of copyrights in AI-generated content. ChatGPT can serve as a valuable tool for research and development, assisting scholars in literature reviews, hypotheses development, conceptualizing complex concepts and providing a route map for study conduction ([Atkinson, 2024](#)). This study also promotes the usage of ChatGPT and prompts considerations about academic integrity, plagiarism and the proper citation of AI-generated content. The emergence of ChatGPT among educational institutions can help students in clarify the basic concepts and technical tasks ([Javaid et al., 2023](#)). It also helps students save time and perceive the topic in a more accurate way, reducing conflicts of opinion and facilitating better understanding. Lastly, ChatGPT can improve library services by providing personalized recommendations, answering user inquiries and facilitating virtual reference services.

5.2 Conclusion

Academic institutions realize the benefits of adopting a digital-first approach, the implementation of sophisticated technologies like chatbots and virtual reality is expanding in prevalence. Concerns do, however, exist regarding the implementation of AI in academic institutions, specifically privacy infringement and unequal access to digital resources and technology. Academic establishments are tasked with striking a delicate equilibrium between infusing students' lives with institution-wide significance and safeguarding their privacy. Online monitoring platforms may also exacerbate socioeconomic disparities by negatively impacting students who lack access to an ideal, peaceful working environment. The COVID-19 pandemic has brought to light the digital divide within educational institutions, as some students are deprived of technology and Internet access. This discrepancy has led to a divergence in educational opportunities and digital literacy. It is of the utmost importance to ensure that the implementation of cutting-edge technologies in educational environments, such as ChatGPT, is conducted in an ethical and transparent manner.

5.3 Future scope

The future scope of ChatGPT is vast and depends on how it is developed and applied. With ongoing advancements in AI technology, we can expect to see many exciting applications for ChatGPT and other language models in the years to come. ChatGPT can serve as a valuable

tool for bridging the gap between theory and practice in social sciences by synthesizing research, providing practical examples, facilitating collaboration, offering tailored recommendations and supporting decision-making processes. Furthermore, ChatGPT has a vast scope and can be applied in various fields to facilitate communities, guide practitioners and facilitate operations.

5.4 Future directions

On the base of findings of the study, we are enabled to present the future direction for AI and ChatGPT. Firstly, future studies explore how AI and ChatGPT can be thoroughly assimilated into educational curricula to enhance both learning and teaching experiences. This assimilation will develop modules that teach students and faculty how to effectively use AI for research, task management and knowledge sharing. Secondly, future studies should be conducted on growing need for the development of comprehensive ethical frameworks that guide the use of AI tools like ChatGPT in education. Furthermore, studies could focus on producing guidelines that ensure responsible AI usage, particularly in areas such as academic integrity, content generation and intellectual property rights. Thirdly, future studies should expand ChatGPT's role in academic integrity. Future research could explore the creation of specialized AI tools that assist researchers in various stages of research studies, from literature review to hypothesis testing and data analysis procedures. Fourthly, the potential future research that could explore how AI-driven virtual assistants can be further developed to provide more advanced library services, such as predictive analytics AI-powered information retrieval systems and for resource recommendations.

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